FIG. 4 — THREE-PARTY ENROLLMENT CEREMONY 400

PARTIES 402

EMPLOYER 404	MEMBER 406	INSURER 408
(Token Authority)	(Mobile Device)	(Attestation Authority)
Has: group config, roster, plan templates	Has: PHI/PII (local only), enrollment key (generated)	Has: NO PHI, NO secrets Sees only walletId hashes

STEP 1 — EMPLOYER: Generate GroupEnrollmentToken 410

GroupEnrollmentToken 412 (8 Fields)

tokenIdHash, groupIdHash, planIdHash, employeeIdHash,
familyIdHash, validUntil, issuerHash, nonce

↓ Token transmitted to Member (email/portal/QR)

STEP 2 — MEMBER: On-Device Wallet Derivation 420

Member Device (PHI never leaves) 422

420a: Generate enrollmentKey (random 2FA secret, NEVER transmitted)
420b: Hash each PHI field: h(cardholderId), h(dob), h(firstName), h(lastName)
420c: walletId = Poseidon.hash([domainTag=0, h(cardholderId), h(dob),
h(firstName), h(lastName), h(enrollmentKey)])
420d: Submit to insurer: { walletId, tokenHash, groupIdHash } (NO PHI)

↓ Only walletId + token reference cross PHI boundary

PHI BOUNDARY — NOTHING BELOW CONTAINS PHI/PII, ONLY POSEIDON HASHES

STEP 3 — INSURER: Register Issue Credential 430

Insurer Processing 432

430a: Verify token signature + check roster (by employeeIdHash, no PHI)
430b: Register walletId in WalletRegistry MerkleTree (walletRegRoot updated)
430c: Set lifecycle EMPTY(0) -> REGISTERED(1) -> ELIGIBLE(2) -> ACTIVE(3)
430d: Mint credential with credentialCommitment (see FIG. 3, ref 302)
430e: Initialize ACCESS_CONTROL tree: EligibilityData(leaf 0),
AccumulatorData(leaf 127, all zeros), PriorAuth entries as needed

↓

Issued Credential 434

credentialCommitment = H(credentialType, forestRoot, issuerWalletId,
subjectWalletId, issuedAt, expiresAt, revoked)
NFC card (optional): RecoveryCard v2 (~500 bytes, AES-256-GCM, NTAG216)

INFORMATION SEPARATION (By Party) 440

Data Element	Employer	Member	Insurer
PHI (name,dob,id)	Roster (plain)	Device (plain)	NEVER
Enrollment Key	NEVER	Device only	NEVER
walletId (hash)	NEVER	Derived on device	Registered
Group/Plan config	Creates	Receives token	Verifies token
Credential	NEVER	Holds	Issues

TOKEN LIFECYCLE 450

GENERATED 452 --[employer creates]--> DISTRIBUTED 454 --[member redeems]--> CONSUMED 456

[expires]---> EXPIRED 458

[employer revokes]--> REVOKED 460